



AI-Powered SOP Chat Assistant - Project Presentation



Executive Summary

An intelligent chat interface that allows users to interact with Standard Operating Procedures (SOPs) and internal documentation using natural language. The system provides instant answers, normalization guidance, contact information, and automated approval workflows for document updates.



Introduction

What is This Project?

This is an **AI-powered conversational interface** for internal documentation that:

- Answers questions about SOPs, normalization rules, and workflows
- Provides instant access to catalog system information
- Automates document update approval processes
- Learns from conversations to improve over time
- Works entirely on local infrastructure for data privacy

How is it Made?

Technology Stack:

- **Frontend:** Single-page HTML/CSS/JavaScript application
- **Backend:** Node.js Express server (Port 3003)
- **AI Engine:** Groq API (cloud) with Ollama (local) fallback
- **Document Store:** Obsidian Markdown vault
- **File Converter:** Separate service (Port 3002) for DOCX/PDF processing

Architecture:

```
graph TD
    User[User Browser] --> Frontend[Frontend (HTML/JS)]
    Frontend --> Backend[Backend Server (Express.js)]
    Backend --> AI[AI Processing (Groq → Ollama fallback)]
    AI --> Vault[Document Vault (Obsidian MD files)]
```

Why Do We Need It?

Problems It Solves:

1. Information Overload

- SOPs scattered across multiple documents
- Hard to find specific information quickly
- Manual search through hundreds of pages

2. Normalization Confusion

- Unclear abbreviation rules (is "blk" correct for black?)
- Inconsistent formatting across teams
- No single source of truth

3. Contact Discovery

- Don't know who to contact for specific issues
- Team ownership unclear
- Escalation paths not documented

4. Document Updates

- Manual approval process
- No tracking of change requests
- Risk of unauthorized modifications

5. Knowledge Loss

- Tribal knowledge not documented
- New team members struggle to find information
- Repeated questions waste time

Benefits:

- ✅ **Instant Answers** - Get information in seconds, not hours
 - ✅ **Always Available** - 24/7 access to documentation
 - ✅ **Consistent Responses** - Same answer every time
 - ✅ **Audit Trail** - All conversations and changes logged
 - ✅ **Approval Workflow** - Automated document update requests
 - ✅ **Learning System** - Improves from user corrections
 - ✅ **Privacy First** - All data stays on local infrastructure
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Capabilities

1. Intelligent Question Answering

What it does:

- Understands natural language questions
- Searches across all SOP documents
- Provides accurate, cited answers
- Distinguishes between work-related and general questions

Examples:

- "Is 'blk' the correct representation for black?" → Searches normalization rules
- "Who handles normalization issues?" → Finds contact information
- "What is a tombstone ASIN?" → Explains from SOP documents

2. Normalization Guidance

What it does:

- Validates abbreviations and formatting
- Provides correct representations
- Cites specific SOP sections
- Explains normalization rules

Examples:

- Input: "blk" → Output: "Should be 'Black' (Starfish Modular Title Quality Audits, Section III)"
- Input: "gps" → Output: "Should be 'GPS' (all caps for acronyms)"

3. Contact & Ownership Discovery

What it does:

- Identifies responsible teams
- Provides POC (Point of Contact) information
- Shows escalation paths
- Maps system ownership

Examples:

- "Who handles CSS issues?" → "CSS Team, POC: fsainula"
- "Reconciliation team contact?" → Full team details with escalation matrix

4. Document Update Workflow

What it does:

- Detects update requests automatically
- Creates approval requests
- Saves proposed changes to temp files
- Tracks approval/rejection status
- Updates original documents after approval

Workflow:

```
User: "Add tombstone section to Starfish SOP"
↓
AI: Generates updated document
↓
System: Creates approval request
↓
Admin: Reviews and approves/rejects
↓
System: Updates original document (if approved)
```

5. File Conversion

What it does:

- Converts DOCX files to Markdown
- Extracts text from PDFs
- Processes images with OCR
- Maintains formatting and structure

Supported Formats:

- Microsoft Word (.docx)
- PDF documents (.pdf)
- Images (.png, .jpg) with OCR

6. Conversation Memory

What it does:

- Remembers context within a conversation
- Learns from user corrections
- Stores important facts
- Builds knowledge base over time

Memory Types:

- **Corrections:** User fixes wrong information
- **Rules:** New policies or procedures
- **Facts:** Important details to remember
- **Definitions:** Term explanations

7. Live Updates

What it does:

- Real-time file change notifications
- Instant vault updates
- Live approval status changes
- Server-Sent Events (SSE) for push updates

8. Multi-File Search

What it does:

- Searches across entire vault
- Ranks results by relevance
- Provides context snippets
- Handles large document collections

9. Chat History

What it does:

- Saves all conversations
- Organizes by date
- Detects SOP topics automatically
- Searchable archive

10. Network Accessibility

What it does:

- Local access (127.0.0.1)
- Wi-Fi network sharing
- VPN access support
- Multiple device support

Limitations

Technical Limitations

1. AI Model Constraints

Issue: Token limits and rate limiting

- **Groq API:** 6,000 requests/day free tier, rate limited
- **Context Window:** Limited to ~8KB of document content per query
- **Response Length:** Capped at 1,024 tokens (~800 words)

Impact:

- Long documents must be truncated
- Complex multi-part questions may hit limits
- Rate limits cause fallback to slower local AI

Mitigation:

- Automatic fallback to Ollama (local)
- Smart context extraction (most relevant sections)
- History trimming (keep only last 4 messages)

2. Performance Issues

Issue: Response time varies significantly

- **Groq (cloud):** 2-5 seconds (fast)
- **Ollama (local):** 10-60 seconds (slow)
- **Large documents:** Additional processing time

Impact:

- User may experience delays
- System may appear to hang
- Timeout after 60 seconds

Mitigation:

- Prioritize Groq for speed
- Show loading indicators
- Implement timeouts
- Reduce context size

3. Memory Constraints

Issue: Ollama requires significant RAM

- **Mistral model:** ~4GB RAM
- **Larger models:** Up to 26GB RAM
- **Multiple concurrent users:** RAM multiplies

Impact:

- May not run on low-spec machines
- Slows down other applications
- Limits concurrent users

Mitigation:

- Use smaller models (orca-mini: 1.3GB)
- Limit concurrent requests

- Cloud-only mode (Groq)

4. Search Accuracy

Issue: Keyword-based search has limitations

- May miss semantic matches
- Requires exact or similar terms
- No understanding of synonyms
- Sensitive to spelling

Impact:

- May not find relevant information
- User must rephrase questions
- False negatives possible

Mitigation:

- Multiple keyword extraction
- Fuzzy matching
- Search multiple documents
- User can select specific files

5. File Format Support

Issue: Limited to specific formats

- **Supported:** Markdown (.md), DOCX, PDF, images
- **Not supported:** Excel, PowerPoint, proprietary formats
- **OCR limitations:** Handwriting, poor quality images

Impact:

- Some documents cannot be processed
- Manual conversion required
- Information may be incomplete

Mitigation:

- Convert unsupported formats manually
- Use high-quality scans for OCR
- Maintain Markdown versions

System Limitations

6. Single-User Architecture

Issue: Not designed for high concurrency

- No user authentication
- No session management
- Shared conversation history
- No access control

Impact:

- Not suitable for large teams without modification
- Privacy concerns in multi-user environment

- No user-specific customization

Mitigation:

- Run separate instances per user
- Add authentication layer (future)
- Use network isolation

7. No Real-Time Collaboration

Issue: Changes not synchronized across users

- Each user has own chat history
- Approval workflow is centralized but not real-time
- No conflict resolution

Impact:

- Users may see outdated information
- Duplicate approval requests possible
- Manual coordination needed

Mitigation:

- Refresh vault regularly
- Check approval logs before requesting
- Communicate with team

8. Approval Workflow Limitations

Issue: Manual admin approval required

- No automatic approval rules
- Single admin credential (hardcoded)
- No approval delegation
- No approval notifications

Impact:

- Delays in document updates
- Bottleneck at admin level
- No audit of who approved what

Mitigation:

- Check approval logs regularly
- Add multiple admin accounts (code change)
- Implement notification system (future)

9. Network Dependency

Issue: Requires network for Groq API

- Internet connection needed for fast responses
- Groq API must be reachable
- Fallback to Ollama if offline

Impact:

- Slower responses when offline

- Complete failure if Ollama not installed
- Network issues cause delays

Mitigation:

- Install Ollama for offline mode
- Use local network only
- Cache common responses (future)

10. Data Privacy

Issue: Groq API sends data to cloud

- Questions and context sent to Groq servers
- Potential data leakage
- Compliance concerns for sensitive data

Impact:

- May violate data policies
- Sensitive information exposed
- Audit trail incomplete

Mitigation:

- Use Ollama-only mode for sensitive data
- Review Groq privacy policy
- Implement data filtering (future)

Operational Limitations

11. Setup Complexity

Issue: Requires technical setup

- Node.js installation
- Ollama installation and model download
- Configuration file editing
- Command-line usage

Impact:

- Non-technical users struggle
- Setup takes 20-30 minutes
- Troubleshooting requires technical knowledge

Mitigation:

- Provide detailed documentation
- Create setup scripts
- Offer pre-configured packages

12. Maintenance Requirements

Issue: Ongoing maintenance needed

- Models need updates
- Dependencies need updates
- Vault files need organization

- Logs need cleanup

Impact:

- System degrades over time
- Security vulnerabilities
- Disk space issues

Mitigation:

- Schedule regular maintenance
- Automate updates (future)
- Monitor disk usage

13. Error Handling

Issue: Limited error recovery

- Cryptic error messages
- No automatic retry for some failures
- Manual intervention required

Impact:

- User frustration
- Lost conversations
- System appears broken

Mitigation:

- Improve error messages
- Add automatic retry logic
- Provide troubleshooting guide

14. Scalability

Issue: Not designed for enterprise scale

- Single server instance
- No load balancing
- No database backend
- File-based storage

Impact:

- Cannot handle hundreds of users
- Performance degrades with large vaults
- No high availability

Mitigation:

- Use for small teams only
- Implement caching (future)
- Consider database migration (future)

15. Documentation Drift

Issue: AI responses based on static documents

- Documents may become outdated

- No automatic update detection
- AI doesn't know when information is stale

Impact:

- Incorrect answers if documents outdated
- User trust eroded
- Manual verification needed

Mitigation:

- Regular document reviews
- Version control for documents
- Add "last updated" timestamps

 Summary Table

Category	Capability	Limitation
AI	Natural language understanding	Token limits, rate limiting
Performance	Fast cloud responses (Groq)	Slow local fallback (Ollama)
Search	Multi-document search	Keyword-based, not semantic
Memory	Conversation context	Limited to 4 messages
Files	DOCX, PDF, MD support	No Excel, PPT support
Users	Single-user optimized	No multi-user auth
Updates	Approval workflow	Manual admin approval
Network	Local + Wi-Fi access	Internet needed for Groq
Privacy	Local Ollama option	Groq sends data to cloud
Setup	Documented process	Requires technical skills
Scale	Small team usage	Not enterprise-ready
Maintenance	Self-contained	Manual updates needed

 Ideal Use Cases

 Perfect For:

- Small teams (5-20 people)
- Internal documentation Q&A
- SOP compliance checking
- Normalization guidance
- Contact discovery
- Document update workflows

 Not Suitable For:

- Enterprise-wide deployment (100+ users)
 - Real-time collaboration
 - Highly sensitive data (without Ollama-only mode)
 - Mission-critical applications
 - Public-facing services
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Future Improvements

Short Term (1-3 months)

- ☐ Reduce system prompt size (50% reduction)
- ☐ Implement better caching
- ☐ Add user authentication
- ☐ Improve error messages
- ☐ Add notification system

Medium Term (3-6 months)

- ☐ Semantic search (embeddings)
- ☐ Multi-user support
- ☐ Real-time collaboration
- ☐ Mobile app
- ☐ Advanced analytics

Long Term (6-12 months)

- ☐ Enterprise features
 - ☐ Database backend
 - ☐ High availability
 - ☐ API for integrations
 - ☐ Advanced AI models
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Contact & Support

Project Owner: [adisba] **Technical Contact:** adisba@amazon.com **Documentation:** See `DOCUMENTATION_INDEX.md` **Setup Guide:** See `SETUP_GUIDE.md`

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